

- **Caffe2DML API:** This API enables the model to be represented in Caffe's proto format.
- **Keras2DML API:** This API enables the model to be represented in Keras.

A code snippet with Keras2DML, to implement ResNet50, is shown below:

```
import os
os.environ['CUDA_DEVICE_ORDER'] = 'PCI_BUS_ID'
os.environ['CUDA_VISIBLE_DEVICES'] = ''

# Set channel first layer
from keras import backend as K
K.set_image_data_format('channels_first')

from systemml.mllearn import Keras2DML
import systemml as sml
import keras, urllib
from PIL import Image
from keras.applications.resnet50 import preprocess_input,
decode_predictions, ResNet50

keras_model = ResNet50(weights='imagenet', include_top=True, pooling='None', input_shape=(3, 224, 224))
keras_model.compile(optimizer='sgd', loss='categorical_crossentropy')

sysml_model = Keras2DML(spark, keras_model, input_shape=(3, 224, 224), weights='weights_dir', labels='https://raw.githubusercontent.com/apache/systemml/master/scripts/nn/examples/caffe2dml/models/imagenet/labels.txt')
sysml_model.summary()
urllib.urlretrieve('https://upload.wikimedia.org/wikipedia/
```

```
commons/f/f4/Cougar_sitting.jpg', 'test.jpg')
img_shape = (3, 224, 224)
input_image = sml.convertImageToNumPyArr(Image.open('test.jpg'), img_shape=img_shape)
sysml_model.predict(input_image)
```

As the SystemML is still evolving, the road map for future features includes enhanced deep learning support, support for distributed GPUs, etc.

To summarise, SystemML aims to position itself as SQL for machine learning. It enables developers to implement and optimise machine learning code with ease and effectiveness. Scalability and performance are its major advantages. The ability to run on top of Spark makes automatic scaling possible. With the planned expansion of deep learning features, SystemML will become stronger in future releases. If you are a machine learning enthusiast, then SystemML is a platform that you should try. **END** 🐧

References

- [1] SystemML Home Page: <https://systemml.apache.org/>
- [2] SystemML Press Release: <https://blogs.apache.org/foundation/entry/the-apache-software-foundation-announces13>
- [3] SystemML Documentation: <http://apache.github.io/systemml/>

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